Abdul Rub Mohammed

Gen-AI Engineer | https://www.linkedin.com/in/abdulrubmohammed/
https://github.com/MohammedAbdulRub | https://mohammedabdulrub.github.io/

PROFESSIONAL SUMMARY

AI Engineer with 3+ years at Infosys building RAG-based GenAI co-pilots, LLM evaluation, and Python/Spark/SQL pipelines on AWS for enterprise insurance clients, plus 2026 internship work on LLM agents. M.S. Data Science candidate at UNT (GPA 3.9, Dec 2026). Ships AI automation into production workflows, cutting insurance document processing from ~1 hour to minutes.

SKILLS

- **Generative AI & LLMs:** RAG, LangChain, LLM Agents, LLM Evaluation, Prompt Engineering, Fine-Tuning, Groq API, Azure OpenAI, Anthropic Claude API.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, LSTM/RNN, Time-Series Regression.
- **Data Engineering:** Python, Java, SQL, C++, API Integration, Automated Testing, Spark, Hive, Hadoop.
- **Cloud & MLOps:** AWS (S3, SageMaker, DynamoDB), Azure ML, GCP, Docker, Git.
- **Analytics & Visualization:** Pandas, NumPy, Tableau, Power BI, Looker, Matplotlib, statsmodels.

EXPERIENCE

Optiu.ai | AI Engineer Intern ***June 2026 – Aug 2026***

- Co-developed LLM agents for sales forecasting and customer retention on the OptiU AI platform, orchestrating Groq-hosted LLMs with time-series regression models to generate plain-language recommendations for carwash and grocery SMBs; piloted with 4 prospective clients.
- Built Python integrations with the Groq API and automated test suites for agentic workflows (optimization, deal-closing) in the OptiU Platform SDK, enabling rapid iteration across agent releases.

University of North Texas | Math Lab Tutor ***Feb 2026 - Present***

- Tutor students one-on-one and in groups in probability & statistics, linear algebra, and calculus; over 80% of tutored students improved their course grades.
- Translate complex quantitative concepts into accessible frameworks, strengthening the analytical foundation directly applicable to ML model development and statistical inference to the students.

Infosys | Associate Data Scientist ***2021 - 2024***

- Co-built a RAG-based GenAI document co-pilot for enterprise insurance clients, engineering Spark/Hive and SQL ingestion pipelines for unstructured documents and integrating validated outputs with SAP via Java workflows.
- Co-designed the LLM evaluation framework for the co-pilot, measuring extraction accuracy and error rates against a ground-truth set of several thousand documents to gate production rollout.
- Developed an ML-assisted insurance document-processing pipeline on AWS (Python, scikit-learn, DynamoDB), cutting processing time from ~1 hour to a few minutes per batch for downstream claims and underwriting teams.
- Automated generation of outgoing insurance correspondence (claim letters and return/refund notices to insureds and policy owners) with Java, saving 8–10 minutes per document across 200+ weekly transactions (~30 staff-hours/week).
- Earned 2 Infosys INSTA Awards as one of 5 engineers delivering AI solutions for international insurance clients, partnering remotely with on-site client leads in Agile sprints and shipping 2 weeks ahead of schedule.
- Streamlined stakeholder reporting with Tableau dashboards on AWS-backed Python and built-in data-quality checks, cutting report production from 1–2 hours to minutes per batch and reducing stakeholder correction requests by 30%.

PROJECTS

- **Human Activity Recognition using LSTM**
  Trained a 2-layer stacked LSTM on raw UCI HAR smartphone sensor data, reaching 90.8% mean test accuracy (93.1% best) versus 86.8–88.2% for CNN baselines; an accelerometer-only ablation dropped accuracy to 87%. **Tech: Python, TensorFlow/Keras, Scikit-Learn.**
- **Scalable Ride-Sharing Database System**
  Designed a normalized relational database schema with constraint-based business rules to optimize end-to-end ride-sharing operations such as booking lifecycle, secure payments, and feedback using analytical SQL queries supporting 10K+ simulated ride transactions. **Tech: SQL, MySQL, RDBMS, Constraint Modeling.**
- **Healthcare Claims Outcomes Analysis**
  Analyzed 1,000+ healthcare claims to profile outcomes, denial reasons, and payment patterns; identified the leading denial reasons and their share of total denials and surfaced disparities in denial rates to inform revenue-cycle prioritization. **Tech: Python, Tableau, Pandas.**

EDUCATION

- **University of North Texas** Master of Science | Data Science | GPA: 3.9 ***Jan 2025 – Dec 2026***
- **Osmania University** Bachelor of Engineering | Computer Science & Engineering